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Mini project 3

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14.1	Topics to choose from	

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You need to calibrate the cameras according to what was covered during the stereo vision classes. Complete **one** of the topics (1–3) below. The final calibration result will be evaluated based on a visualization. No presentation is required.

1. **Calibration of RGB and thermal data [5 points]**
 - (a) single-camera calibration,
 - (b) stereo calibration,
 - (c) image overlay (registration): pixel (x, y) with color X corresponds to temperature Y .
2. **Calibration of a real stereo rig (hardware) [10 points]**
 - (a) start the cameras and install drivers,
 - (b) record a set of checkerboard images,
 - (c) calibrate each camera and perform stereo calibration.
3. **Calibration of event-based and RGB data [15 points]**
 - (a) a Jupyter Notebook with instructions is provided,
 - (b) calibration data for RGB and event camera are provided in separate files,
 - (c) perform stereo calibration and overlay the streams,
 - (d) additionally, use appropriate functions to detect a face.